

Comparison of CNN and MobileNetV2 for Quality Classification of Orange, Apple, and Lemon Fruits

Nurhikmah¹

¹ Computer Science, Computer System, University Dinamika Bangsa, Jambi, Indonesia

Email: nurhikmah25924@gmail.com

Corresponding Author Email : nurhikmah25924@gmail.com

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Abstrak—Background: Fruit quality classification plays an important role in supporting quality control processes in the food industry. Manual inspection is often time-consuming, inconsistent, and highly dependent on human observation. Therefore, this study compares the performance of a Convolutional Neural Network (CNN) developed from scratch and a MobileNetV2 Transfer Learning model with fine-tuning for automatic fruit quality classification. The dataset consists of six classes, namely fresh apples, fresh lemons, fresh oranges, rotten apples, rotten lemons, and rotten oranges. Before training, all images were resized and augmented to improve model generalization. The CNN from Scratch model was trained using a custom convolutional architecture, while MobileNetV2 utilized pre-trained ImageNet weights followed by fine-tuning. Model performance was evaluated using training accuracy, validation accuracy, testing accuracy, confusion matrix, precision, recall, and F1-score. Experimental results show that the CNN from Scratch model achieved **99.02% training accuracy, 99.27% validation accuracy, and 99.18% testing accuracy**, whereas MobileNetV2 achieved **99.89% training accuracy, 99.63% validation accuracy, and 99.27% testing accuracy**. Both models demonstrated excellent classification performance, but MobileNetV2 consistently produced slightly higher accuracy and better generalization. These findings indicate that transfer learning with MobileNetV2 is a more effective approach for fruit quality classification than designing a CNN model from scratch.

Keywords: *Fruit Quality Classification; Convolutional Neural Network; MobileNetV2; Transfer Learning; Deep Learning; Image Classification*

1. INTRODUCTION

The rapid advancement of **Artificial Intelligence (AI)**, particularly **Deep Learning**, has made significant contributions to the field of digital image processing and **computer vision**. One of its most prominent applications is the automatic classification of fruit quality using digital images. Automated classification systems can assist the fruit sorting process based on freshness levels, thereby improving efficiency, consistency, and accuracy while reducing errors caused by manual inspection. Such technology has become increasingly important in agriculture, food distribution, and modern retail industries, where fast and accurate fruit quality assessment is essential for maintaining product quality and customer satisfaction [1].

Traditionally, fruit quality classification has been performed through manual visual inspection by human operators. However, this approach has several limitations, including dependency on the operator's experience, lighting conditions, fatigue, and relatively long inspection times. Consequently, manual inspection often produces inconsistent results, which may affect product quality and the overall efficiency of the distribution process [2]. Therefore, an automated classification system capable of objectively identifying fruit conditions with high accuracy is highly desirable.

In recent years, **Convolutional Neural Networks (CNNs)** have become one of the most widely adopted Deep Learning techniques for image classification tasks. CNNs are capable of automatically extracting hierarchical image features through convolution operations, enabling superior performance compared with conventional feature extraction methods. A **CNN developed from scratch** allows researchers to design a customized network architecture according to the characteristics of the dataset. However, this approach generally requires a large amount of training data and longer training time to achieve optimal performance [3].

Besides developing a CNN from scratch, **Transfer Learning** has emerged as an effective alternative by utilizing models that have been pre-trained on large-scale datasets such as **ImageNet**. Among various transfer learning architectures, **MobileNetV2** has gained considerable attention due to its lightweight architecture, fewer trainable parameters, and high classification accuracy. Furthermore, the **fine-tuning** process enables the pre-trained model to adapt its parameters to a new dataset, resulting in improved generalization performance without requiring the entire network to be trained from the beginning [4].

Several previous studies have demonstrated the effectiveness of both CNN and Transfer Learning in fruit classification tasks. Mureşan and Oltean showed that CNN achieved excellent performance in recognizing multiple fruit categories using the Fruits-360 dataset [5]. Other studies employing MobileNetV2 reported that transfer learning significantly improved classification accuracy while reducing computational cost and training time compared with CNN models trained from scratch [6]. Moreover, comparative studies involving modern CNN

architectures, such as ResNet, EfficientNet, and MobileNet, have indicated that transfer learning generally provides better performance, particularly when the available training data are limited [7]. More recent research has also shown that fine-tuning further enhances classification accuracy without substantially increasing model complexity [8].

Despite these promising findings, most previous studies have primarily focused on achieving the highest classification accuracy without conducting a comprehensive comparison between **CNN From Scratch** and **Transfer Learning** under identical experimental conditions. Furthermore, limited attention has been given to analyzing the generalization capability of these models using training and validation curves, confusion matrices, and evaluation metrics such as precision, recall, and F1-score. This limitation reveals a research gap regarding the comprehensive evaluation of both approaches for fruit quality classification based on freshness conditions.

Based on this research gap, the present study aims to compare the performance of a **CNN From Scratch** model with a **MobileNetV2 Transfer Learning model enhanced through fine-tuning** for fruit quality classification. The dataset consists of six categories, namely **fresh apples, fresh lemons, fresh oranges, rotten apples, rotten lemons, and rotten oranges**. Before model training, all images underwent preprocessing procedures, including image resizing and data augmentation. Model performance was evaluated using **training accuracy, validation accuracy, testing accuracy, confusion matrix, precision, recall, F1-score**, as well as **training and validation loss and accuracy curves**. Experimental results indicate that the CNN From Scratch model achieved **99.02% training accuracy, 99.27% validation accuracy, and 99.18% testing accuracy**, whereas the **MobileNetV2 Transfer Learning with Fine-Tuning** model achieved **99.89% training accuracy, 99.63% validation accuracy, and 99.27% testing accuracy**. These results demonstrate that both models achieved excellent classification performance; however, MobileNetV2 exhibited slightly better generalization capability, making it a more effective solution for automated fruit quality classification.

The primary contribution of this research lies in the comprehensive evaluation of two Deep Learning approaches, namely **CNN From Scratch** and **MobileNetV2 Transfer Learning with Fine-Tuning**, using the same dataset, identical preprocessing procedures, and consistent evaluation metrics. Unlike previous studies that mainly focused on overall classification accuracy, this research also analyzes model generalization through training and validation loss curves, accuracy curves, confusion matrices, and performance metrics including precision, recall, and F1-score. Therefore, the findings are expected to provide valuable insights for researchers and practitioners in selecting the most suitable image classification approach, particularly for computer vision-based fruit quality assessment. Furthermore, this study highlights the trade-off between model complexity, training time, and classification performance, which can serve as a practical reference for developing automated fruit quality inspection systems in agriculture and the food industry.

2. METHODOLOGY

This study employed an experimental research approach to compare the performance of two deep learning models, namely a **Convolutional Neural Network (CNN) developed from scratch** and **MobileNetV2 using Transfer Learning with Fine-Tuning**, for fruit quality classification. The research workflow consisted of dataset collection, data preprocessing, model development, model training, performance evaluation, and comparative analysis. Both models were trained and evaluated using the same dataset, preprocessing techniques, training strategy, and evaluation metrics to ensure a fair comparison. The experiments were conducted using the TensorFlow-Keras framework in the Google Colaboratory environment with GPU acceleration.

A. Dataset Distribution

The dataset used in this research consists of six fruit quality categories: **fresh apples, fresh lemons, fresh oranges, rotten apples, rotten lemons, and rotten oranges**. Each image was manually labeled according to the freshness condition of the fruit. Prior to the training process, all images were inspected to ensure that the dataset was free from corrupted files and incorrect labels.

Table 1 presents the distribution of images in each class. As shown in the table, the dataset is moderately imbalanced because several classes contain substantially more samples than others. Therefore, data augmentation techniques were applied during the training phase to improve the model's generalization capability and reduce the effect of class imbalance.

Table 1. Dataset Distribution

No.	Class	Number of Images
1	Fresh Apples	2,089
2	Fresh Lemons	1,129
3	Fresh Oranges	1,827
4	Rotten Apples	2,941
5	Rotten Lemons	949

No.	Class	Number of Images
6	Rotten Oranges	1,997
Total		10,932

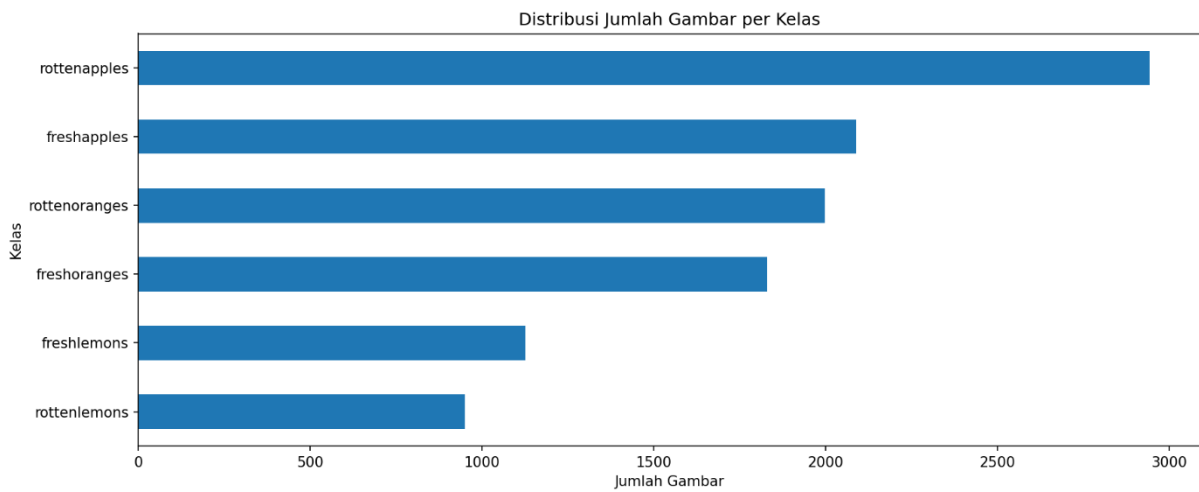


Figure 1. Distribution Class

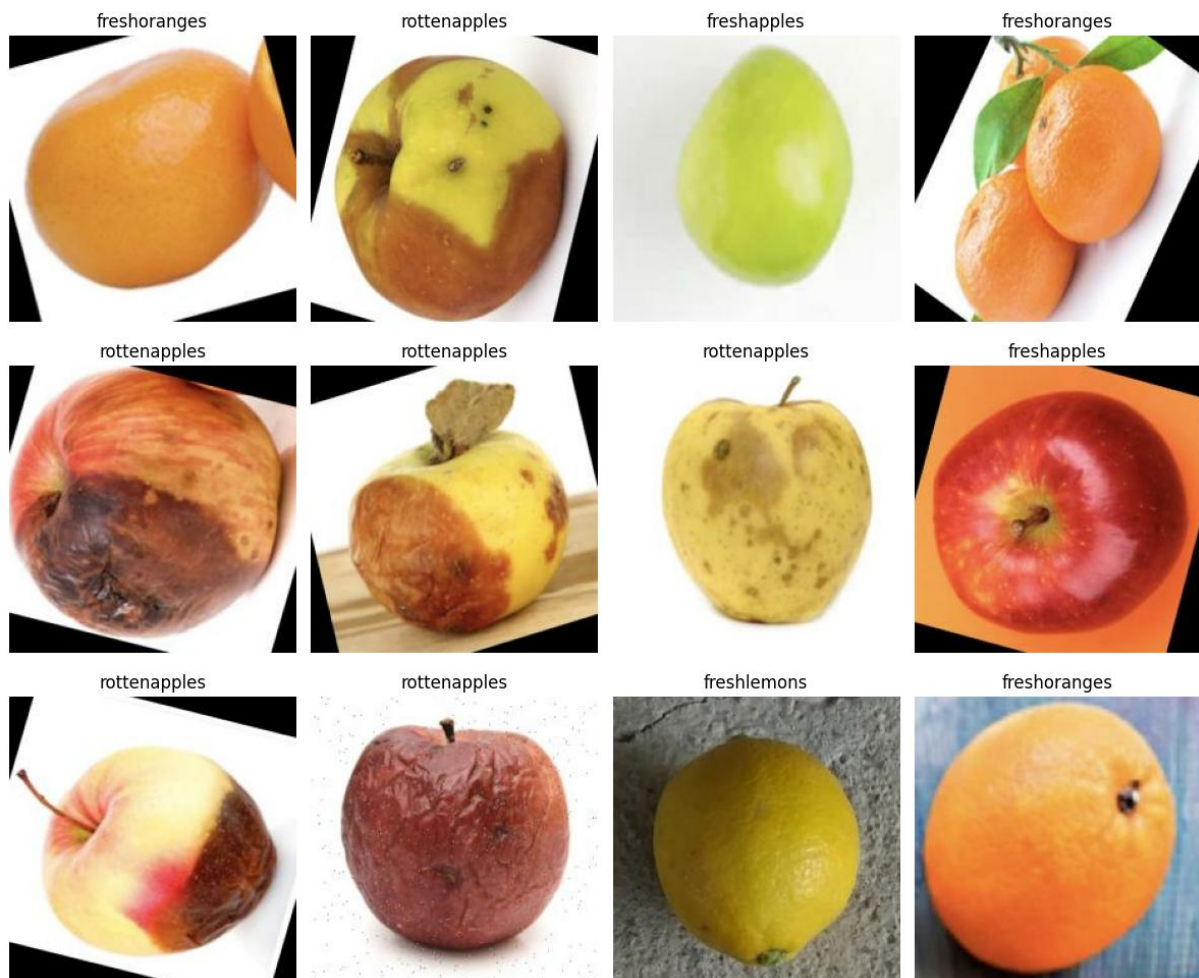


Figure 2. Dataset

B. Data Preparation

The data preparation stage was performed to ensure that all input images were suitable for deep learning training. Initially, every image was resized to **224 × 224 pixels**, which corresponds to the default input size required by MobileNetV2 and also provides a consistent input dimension for the CNN developed from scratch. Pixel values

were subsequently normalized to the range of **0–1** by dividing each pixel intensity by 255, allowing faster convergence during the optimization process.

To improve the robustness and generalization capability of both models, **data augmentation** was applied exclusively to the training dataset. The augmentation process included random rotation, horizontal flipping, zooming, width shifting, and height shifting. These transformations increase the diversity of the training samples without altering the semantic information contained in the images. In contrast, the validation and testing datasets were only resized and normalized without augmentation to ensure an unbiased evaluation of model performance.

After preprocessing, the dataset was divided into three subsets: **training**, **validation**, and **testing**. The training subset was used to optimize the model parameters, the validation subset was employed to monitor learning progress and prevent overfitting, and the testing subset was reserved exclusively for the final performance evaluation.

C. CNN From Scratch Architecture

A custom Convolutional Neural Network (CNN) was developed from scratch to perform fruit quality classification. Unlike transfer learning approaches, the proposed model was initialized using randomly generated weights and learned feature representations directly from the training dataset. The architecture was designed to progressively extract low-level and high-level visual features through multiple convolutional blocks while maintaining computational efficiency.

The input images, with a resolution of $224 \times 224 \times 3$, first passed through a **data augmentation** layer to increase image diversity during training, followed by a **rescaling** layer that normalized pixel values to the range of **0–1**. The feature extraction stage consisted of five convolutional blocks. Each block contained a **Conv2D** layer, **Batch Normalization**, **ReLU activation**, and **MaxPooling2D**. The number of convolutional filters increased gradually from **32, 64, 128, 256**, to **384**, allowing the network to learn increasingly complex visual patterns while reducing the spatial dimensions of the feature maps.

Instead of using a Flatten layer, the proposed architecture employed **Global Average Pooling (GAP)** to transform the extracted feature maps into a one-dimensional feature vector. This approach significantly reduced the number of trainable parameters and helped minimize overfitting. The extracted features were then processed by a fully connected layer containing **256 neurons**, followed by another **Batch Normalization** layer and a **Dropout** layer to further improve model generalization. Finally, a **Softmax** output layer with **six neurons** generated the probability distribution corresponding to the six fruit quality classes.

The proposed CNN architecture contains **1,377,254 total parameters**, of which **1,375,014** are trainable and **2,240** are non-trainable. The complete architecture is illustrated in **Figure 3**.

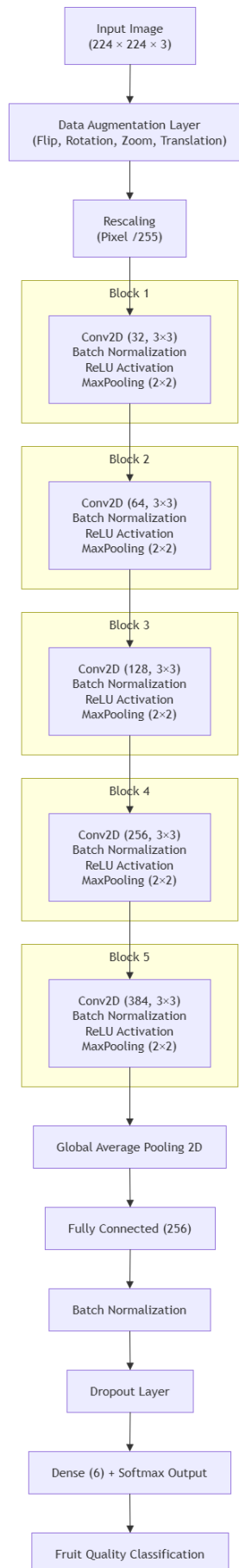


Figure 3. Architecture of the Proposed CNN From Scratch Model

Table 2. Configuration of the Proposed CNN From Scratch

Layer	Configuration
Input Size	$224 \times 224 \times 3$
Data Augmentation	Random Flip, Rotation, Zoom
Rescaling	Pixel value normalization (1/255)
Conv Block 1	Conv2D (32) + BN + ReLU + MaxPooling
Conv Block 2	Conv2D (64) + BN + ReLU + MaxPooling
Conv Block 3	Conv2D (128) + BN + ReLU + MaxPooling
Conv Block 4	Conv2D (256) + BN + ReLU + MaxPooling
Conv Block 5	Conv2D (384) + BN + ReLU + MaxPooling
Feature Aggregation	Global Average Pooling
Fully Connected	Dense (256)
Regularization	Batch Normalization + Dropout
Output Layer	Dense (6, Softmax)
Total Parameters	1,377,254
Trainable Parameters	1,375,014
Model Size	5.25 MB

D. MobileNetV2 Transfer Learning Architecture

To evaluate the effectiveness of transfer learning, this study employed **MobileNetV2**, a lightweight convolutional neural network pre-trained on the **ImageNet** dataset. MobileNetV2 is widely used for image classification because it combines high classification performance with low computational complexity through the use of **depthwise separable convolutions** and **inverted residual bottleneck blocks**. These characteristics make the architecture suitable for deployment on devices with limited computational resources while maintaining competitive classification accuracy.

In this study, the original MobileNetV2 classification head was removed by setting `include_top=False`, allowing the network to function as a feature extractor. The pre-trained convolutional base consists of **154 layers** and was initialized using ImageNet weights. During the initial training stage, all layers of the base model were **frozen** (`trainable=False`), meaning that the pre-trained weights were preserved while only the newly added classification layers were updated. This strategy enables the model to exploit previously learned visual features without requiring the entire network to be retrained, thereby reducing computational cost and training time.

The input images with a resolution of $224 \times 224 \times 3$ first passed through a **data augmentation** layer, followed by the MobileNetV2 preprocessing layer that normalizes the input according to the requirements of the pre-trained model. Feature maps generated by MobileNetV2 were then converted into a one-dimensional feature vector using a **Global Average Pooling** layer. The extracted features were subsequently processed by a fully connected layer containing **256 neurons**, followed by **Batch Normalization** and **Dropout** layers to improve training stability and reduce overfitting. Finally, a **Softmax** output layer with **six neurons** produced the probability distribution corresponding to the six fruit quality categories.

The proposed MobileNetV2 Transfer Learning model contains **2,588,486 total parameters**, including **329,990 trainable parameters** and **2,258,496 non-trainable parameters**. The complete architecture of the proposed model is illustrated in **Figure 4**.

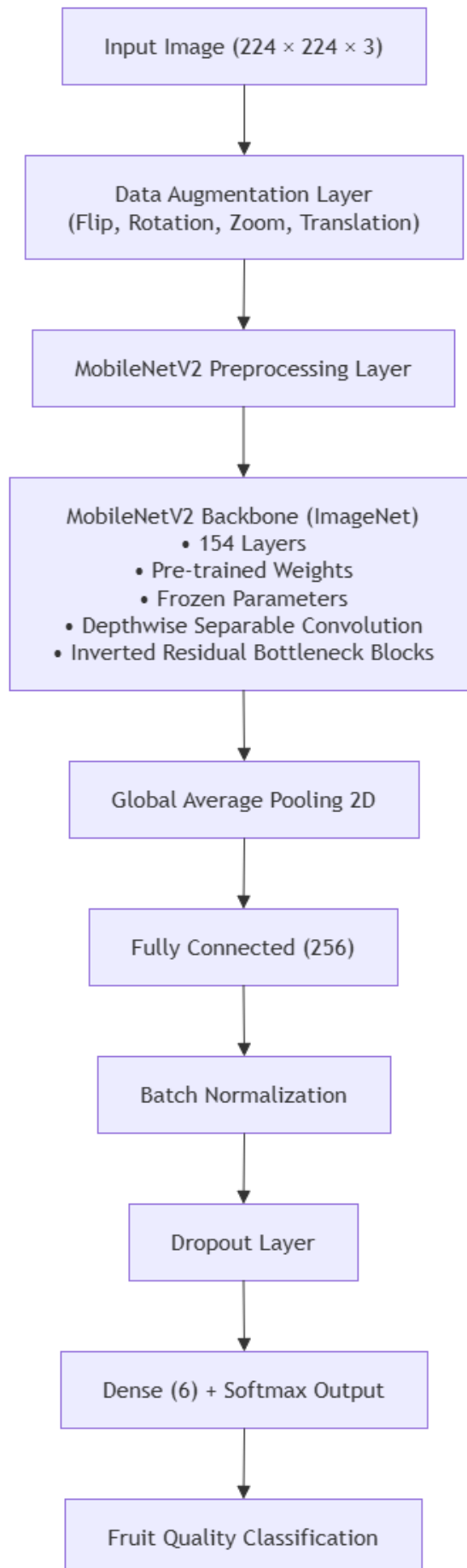


Figure 4. Architecture of the Proposed MobileNetV2 Transfer Learning Model

Table 3. Configuration of the Proposed MobileNetV2 Transfer Learning Model

Layer	Configuration
Input Size	$224 \times 224 \times 3$
Data Augmentation	Random Flip, Rotation, Zoom
Preprocessing	MobileNetV2 Rescaling
Backbone Network	MobileNetV2 (ImageNet)
Number of Backbone Layers	154
Base Model	Frozen (trainable=False)
Feature Aggregation	Global Average Pooling
Fully Connected	Dense (256)
Regularization	Batch Normalization + Dropout
Output Layer	Dense (6, Softmax)
Number of Classes	6
Total Parameters	2,588,486
Trainable Parameters	329,990
Non-trainable Parameters	2,258,496
Model Size	9.87 MB

E. Evaluation Metrics

The performance of both classification models was evaluated using several standard metrics for multi-class image classification. These metrics provide comprehensive information regarding the prediction capability, classification accuracy, and generalization performance of each model. The evaluation was performed on the testing dataset using the confusion matrix and classification report.

1) Accuracy

Accuracy measures the proportion of correctly classified samples to the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- **TP** = True Positive
- **TN** = True Negative
- **FP** = False Positive
- **FN** = False Negative

Accuracy provides an overall indication of the model's classification performance.

2) Precision

Precision measures the proportion of correctly predicted positive samples among all predicted positive samples.

$$Precision = \frac{TP}{TP + FP}$$

A high precision value indicates that the model produces a low number of false positive predictions.

3) Recall

Recall, also known as sensitivity, measures the proportion of actual positive samples that are correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Higher recall indicates that the model successfully detects most of the positive samples.

4) F1-Score

The F1-score is the harmonic mean of Precision and Recall, providing a balanced evaluation of both metrics.

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

A higher F1-score indicates better overall classification performance, especially for datasets with class imbalance.

5) Categorical Cross-Entropy Loss

Since this study addresses a multi-class classification problem, the **Categorical Cross-Entropy** loss function was employed during model training. This loss function measures the difference between the predicted probability distribution and the true class labels.

$$Loss = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where:

- C = number of classes
- y_i = ground-truth label
- $\hat{y}_i$ = predicted probability

Lower loss values indicate better model convergence during training.

6) Confusion Matrix

The confusion matrix provides a detailed visualization of the classification results by comparing predicted labels with actual labels. Each row represents the actual class, whereas each column represents the predicted class. The confusion matrix enables the identification of correctly classified samples as well as misclassified instances for every fruit category.

7) Classification Report

In addition to the confusion matrix, a classification report was generated to summarize the values of **Precision**, **Recall**, **F1-score**, and **Support** for each fruit quality class. This report provides a comprehensive evaluation of the classification performance across all categories and facilitates comparison between the CNN From Scratch and MobileNetV2 Transfer Learning models.

3. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the CNN From Scratch and MobileNetV2 Transfer Learning models. Both models were trained using the same dataset, preprocessing techniques, and evaluation metrics to ensure a fair comparison. The performance of each model was analyzed based on the learning curves, confusion matrices, classification reports, and quantitative evaluation metrics. Furthermore, a comparative analysis was conducted to identify the advantages and limitations of each approach in fruit quality classification.

A. Training Results

The training process was evaluated by monitoring the accuracy and loss values on both the training and validation datasets. These learning curves provide insight into the convergence behavior and generalization capability of each model.

1) CNN From Scratch

The CNN From Scratch model demonstrated stable convergence throughout the training process. As shown in **Figure 5**, both the training and validation accuracy increased steadily and reached values above **99%**, indicating that the model successfully learned representative features from the dataset. The corresponding loss curves shown in **Figure 6** consistently decreased and stabilized near zero without significant fluctuations, suggesting that the optimization process converged successfully.

The final performance of the CNN model achieved **99.02% training accuracy**, **99.27% validation accuracy**, and **99.18% testing accuracy**. The small difference between the training and validation accuracy indicates that the proposed CNN architecture generalized well to unseen data and experienced minimal overfitting.

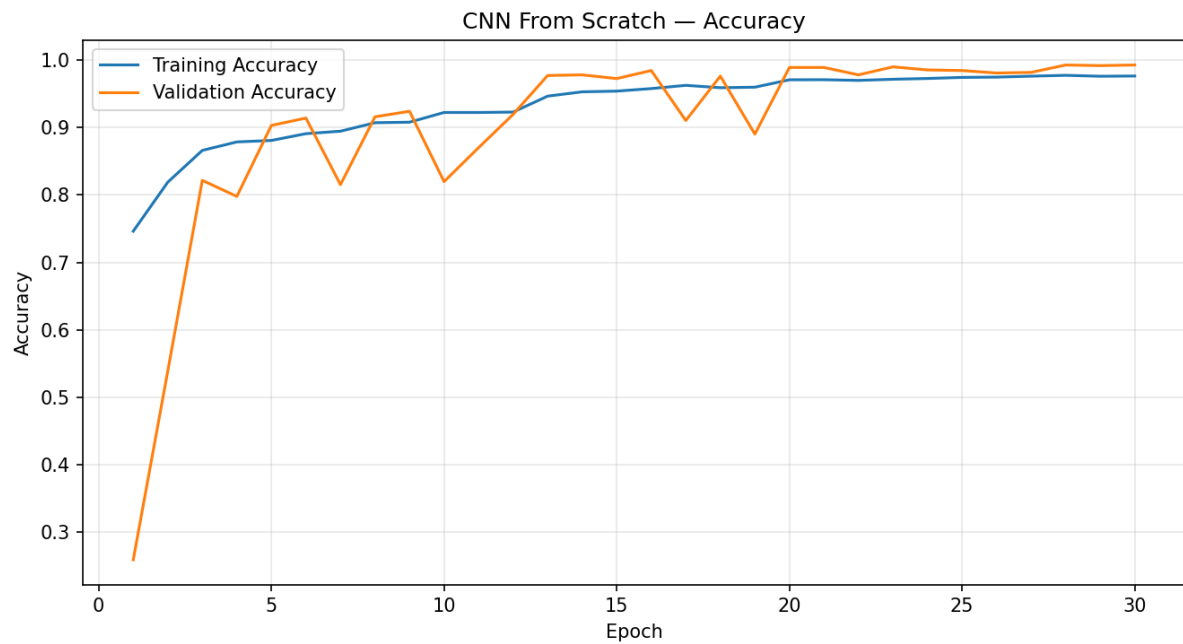


Figure 5. Training and Validation Accuracy of CNN From Scratch

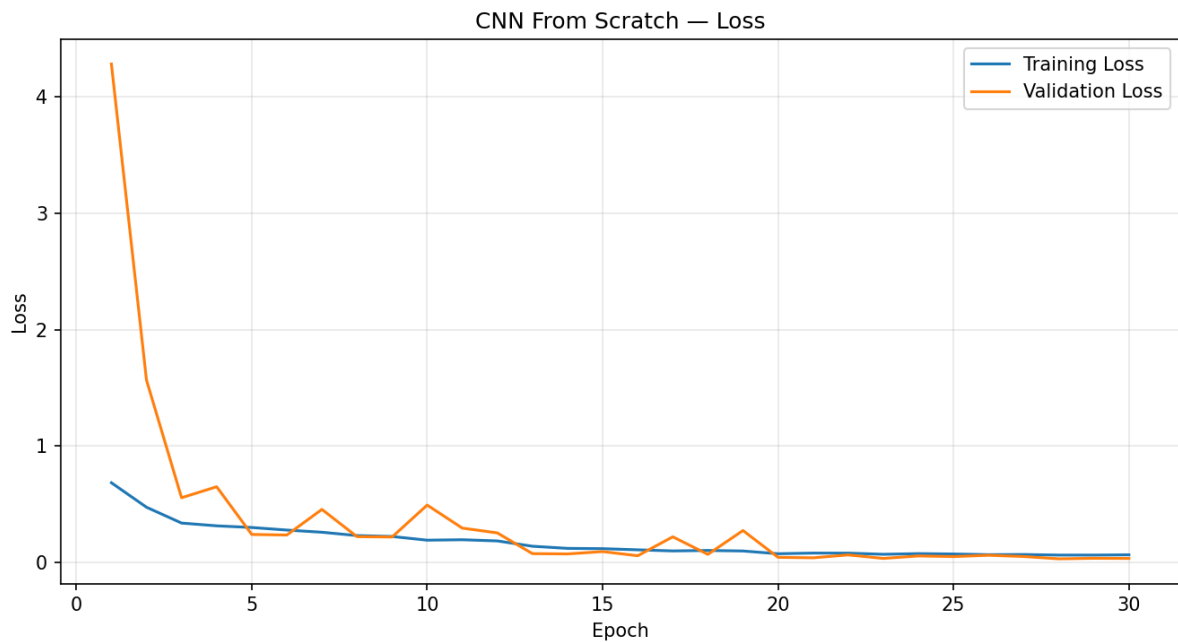


Figure 6. Training and Validation Loss of CNN From Scratch

2) MobileNetV2 Transfer Learning

The MobileNetV2 Transfer Learning model also exhibited excellent learning performance. As illustrated in Figure 7, both training and validation accuracy rapidly converged to values above 99% after several training epochs. Compared with the CNN From Scratch model, MobileNetV2 achieved slightly higher training accuracy while maintaining stable validation performance.

The loss curves presented in Figure 8 show that both training and validation loss gradually decreased and remained close to zero at the end of training. This behavior indicates that the pretrained MobileNetV2 backbone effectively extracted discriminative image features, allowing the classification head to converge efficiently.

The MobileNetV2 model achieved **99.89% training accuracy**, **99.63% validation accuracy**, and **99.27% testing accuracy**, demonstrating slightly superior performance compared with the CNN From Scratch model.

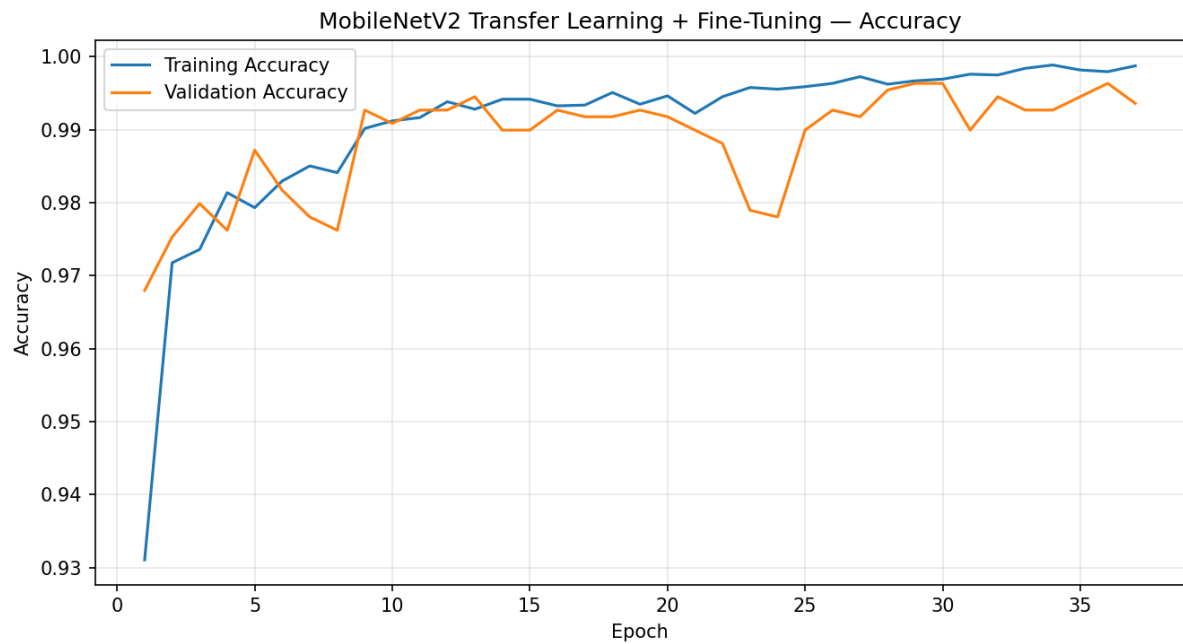


Figure 7. Training and Validation Accuracy of MobileNetV2

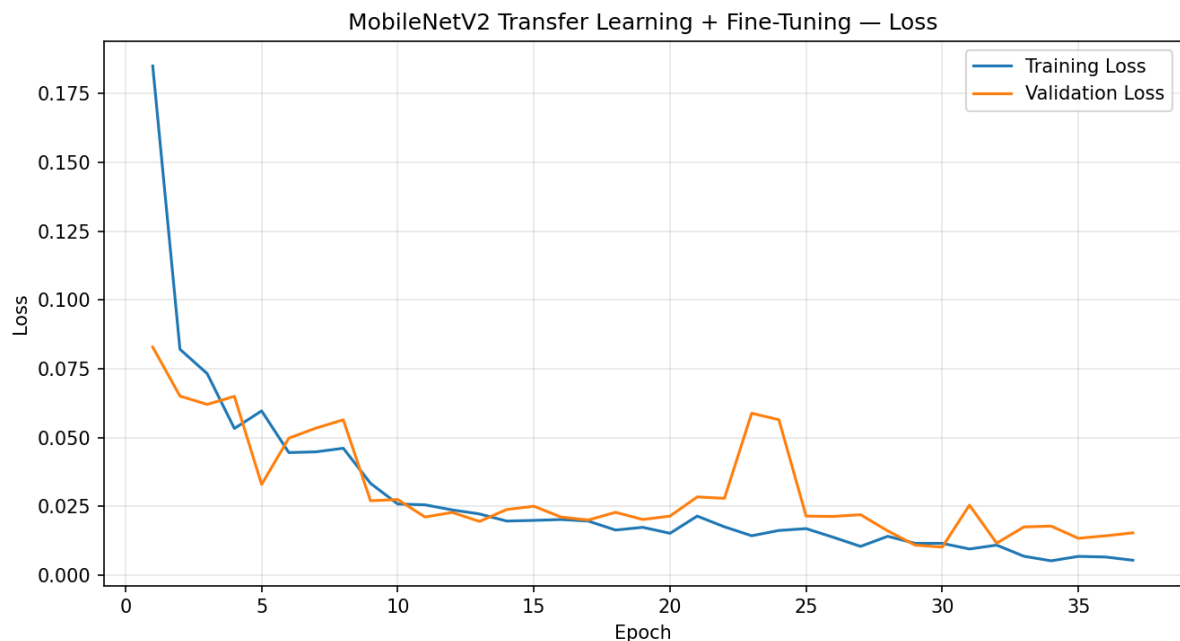


Figure 8. Training and Validation Loss of MobileNetV2

B. Testing Results

To evaluate the classification capability of both models, the trained networks were tested using an independent testing dataset. The evaluation was performed using confusion matrices and classification reports.

1) CNN From Scratch

The confusion matrix shown in **Figure 9** indicates that the CNN From Scratch model correctly classified almost all testing samples. Most predictions are concentrated along the main diagonal, demonstrating excellent classification performance for all six fruit quality categories. Only a small number of misclassified samples were observed, suggesting that the proposed CNN architecture successfully learned discriminative visual features.

The classification report further confirms this observation, where the precision, recall, and F1-score values for all classes were close to **1.00**, indicating balanced classification performance across the dataset.

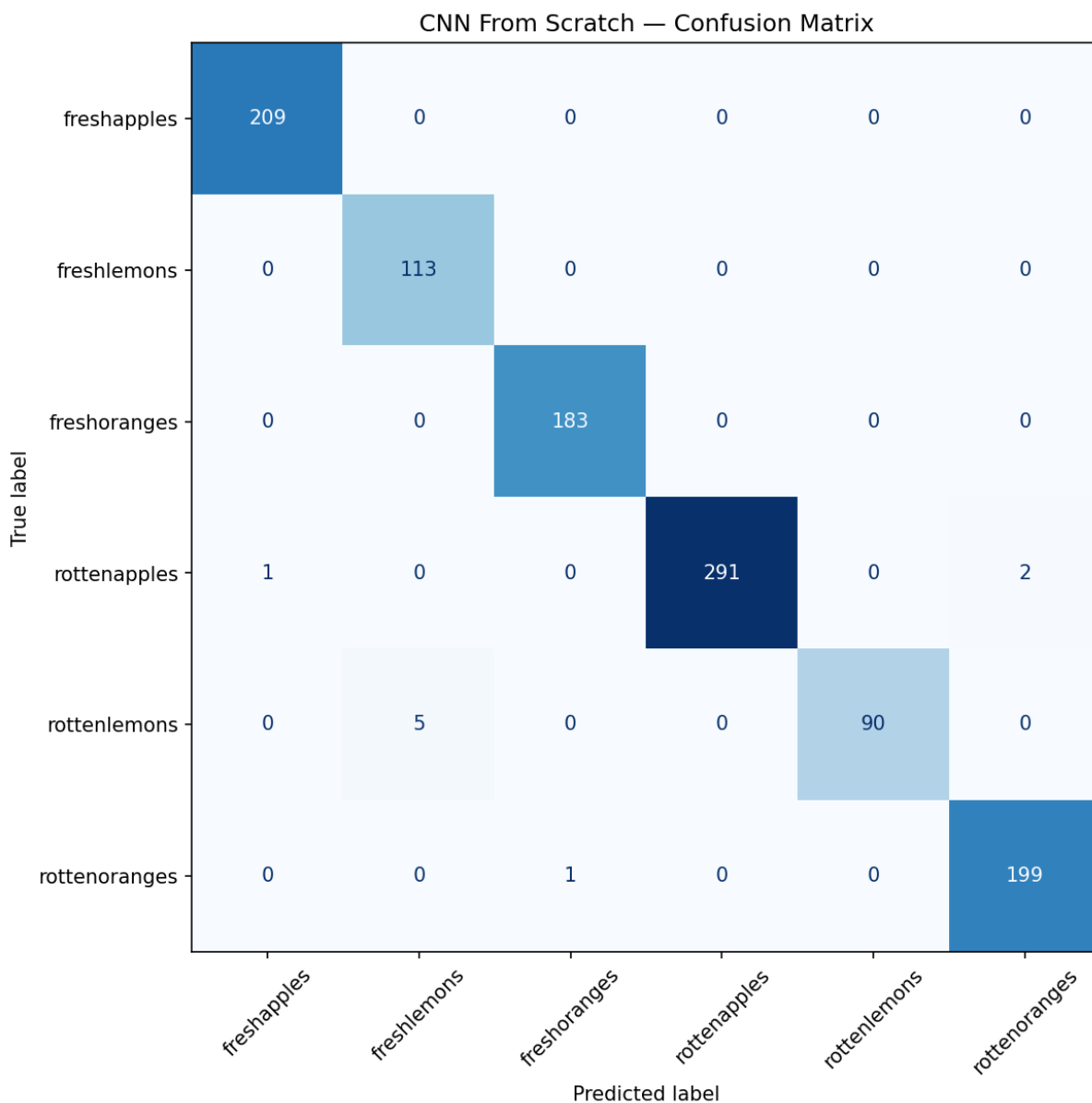


Figure 9. Confusion Matrix of CNN From Scratch

2) MobileNetV2 Transfer Learning

The confusion matrix presented in **Figure 10** demonstrates that MobileNetV2 correctly classified nearly every testing image. Similar to the CNN model, the majority of predictions appear along the diagonal of the matrix, while very few classification errors are observed.

The corresponding classification report shows consistently high precision, recall, and F1-score values across all six fruit classes. These results indicate that MobileNetV2 provides robust feature extraction capability and excellent generalization performance for fruit quality classification.

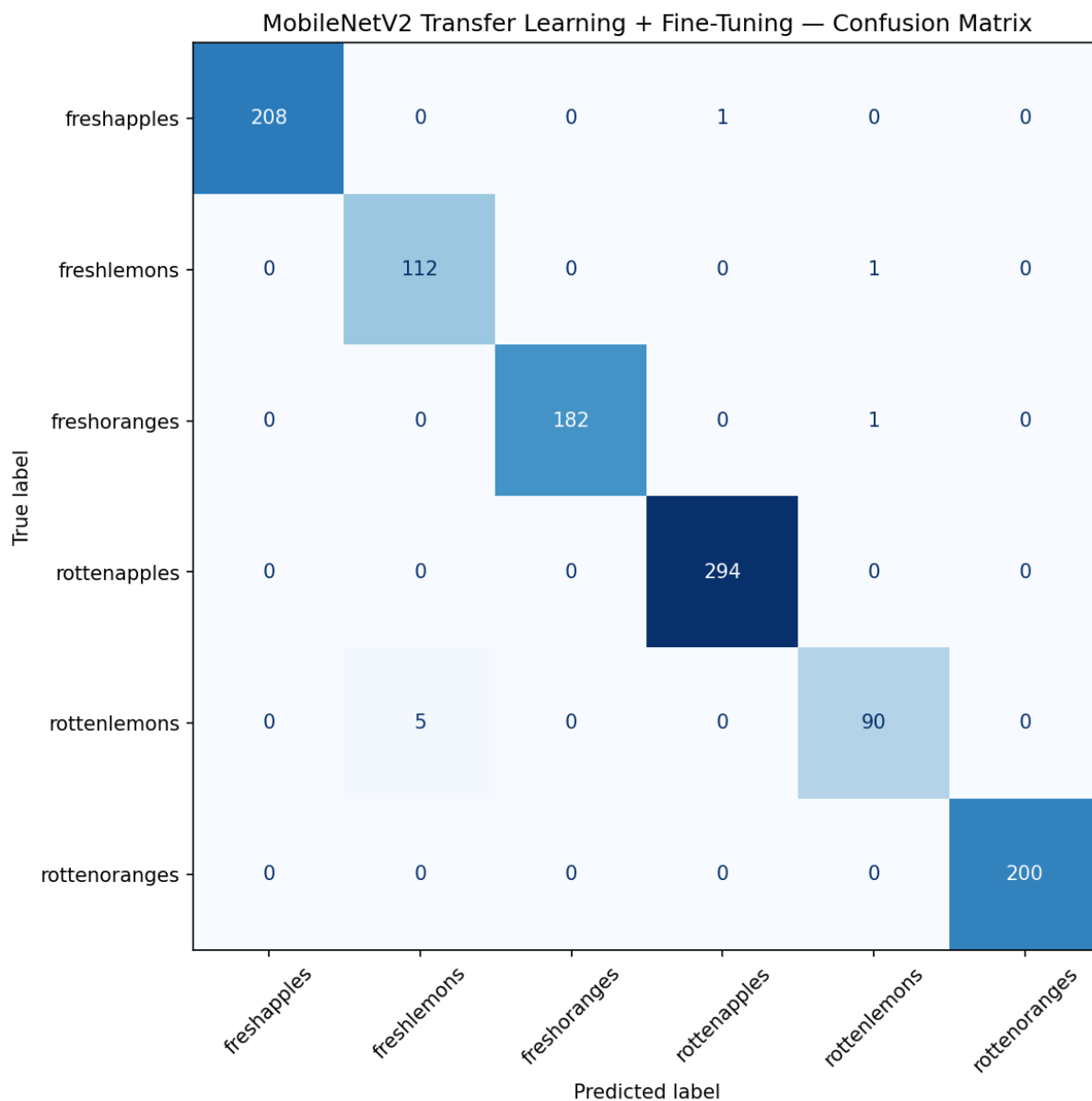


Figure 10. Confusion Matrix of MobileNetV2 Transfer Learning

C. Comparative Analysis

Table 4 summarizes the performance comparison between the CNN From Scratch and MobileNetV2 Transfer Learning models.

Table 4. Performance Comparison of CNN From Scratch and MobileNetV2

Parameter	CNN From Scratch	MobileNetV2 Transfer Learning
Total Parameters	1,377,254	2,588,486
Trainable Parameters	1,375,014	329,990
Model Size	5.25 MB	9.87 MB
Training Accuracy	99.02%	99.89%
Validation Accuracy	99.27%	99.63%
Testing Accuracy	99.18%	99.27%
Generalization	Good	Excellent

The comparison indicates that both models achieved excellent classification performance with testing accuracy exceeding 99%. Although the CNN From Scratch model contained fewer total parameters and a smaller model size, MobileNetV2 consistently produced higher training, validation, and testing accuracy. This improvement can

be attributed to the use of pretrained ImageNet features, which provide richer feature representations than those learned solely from the training dataset.

Another important observation is that MobileNetV2 required only **329,990 trainable parameters**, significantly fewer than the CNN From Scratch model. Consequently, transfer learning reduced the optimization complexity while maintaining superior classification performance. These findings demonstrate that pretrained feature extractors can effectively improve model accuracy without requiring the entire network to be trained from scratch. Overall, the experimental results suggest that MobileNetV2 offers a better trade-off between classification performance and training efficiency, making it a suitable solution for practical fruit quality classification applications.

D. Deployment and Inference Results

To validate the practical applicability of the proposed models, inference was performed using unseen fruit images. During deployment, an input image first undergoes preprocessing before being passed to the trained classification model. The model then predicts one of the six predefined fruit quality categories along with the corresponding confidence score.

The deployment results demonstrate that both CNN From Scratch and MobileNetV2 successfully classified fruit images with high confidence. However, MobileNetV2 generally produced slightly higher confidence scores and more stable predictions, indicating stronger feature representation and better generalization capability. These results confirm that the proposed models can be effectively deployed as an automated fruit quality classification system for real-world agricultural and food quality inspection applications.

4. CONCLUSION

This study presented a comparative evaluation between a **CNN From Scratch** model and a **MobileNetV2 Transfer Learning** model for fruit quality classification using a six-class image dataset consisting of fresh and rotten apples, lemons, and oranges. Both models were trained and evaluated under identical experimental conditions, including the same dataset, preprocessing techniques, and evaluation metrics, to ensure a fair performance comparison.

The experimental results demonstrated that both approaches achieved excellent classification performance, with testing accuracies exceeding 99%. The CNN From Scratch model achieved **99.02% training accuracy**, **99.27% validation accuracy**, and **99.18% testing accuracy**, indicating that the proposed architecture was capable of learning discriminative visual features directly from the dataset. Meanwhile, the MobileNetV2 Transfer Learning model achieved **99.89% training accuracy**, **99.63% validation accuracy**, and **99.27% testing accuracy**, slightly outperforming the CNN model in all evaluation metrics. The confusion matrices and classification reports further confirmed that both models correctly classified almost all testing samples, demonstrating excellent generalization capability and robust performance across all fruit quality categories.

The comparative analysis indicates that transfer learning provides a more effective solution for fruit quality classification by leveraging pre-trained visual features while requiring substantially fewer trainable parameters than a CNN developed entirely from scratch. Although the CNN From Scratch model produced highly competitive results, MobileNetV2 consistently delivered better classification performance and greater training efficiency. These findings suggest that MobileNetV2 is a more suitable architecture for practical fruit quality inspection systems, particularly in applications that require high accuracy and efficient computational performance.

Future research may extend this work by evaluating more recent lightweight deep learning architectures, such as **EfficientNet**, **EfficientNetV2**, **ConvNeXt**, or **Vision Transformer (ViT)**, and by investigating model compression techniques for deployment on embedded and edge computing devices. In addition, expanding the dataset with more fruit varieties, diverse environmental conditions, and varying illumination settings would improve model robustness and generalization in real-world agricultural and food quality inspection scenarios.

REFERENCES

- [1] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, dan L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," dalam *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, hlm. 4510–4520. doi: 10.1109/CVPR.2018.00474.
- [2] H. Mureşan dan M. Oltean, "Fruit Recognition from Images Using Deep Learning," *Acta Univ. Sapientiae Inform.*, vol. 10, no. 1, hlm. 26–42, 2018, doi: 10.2478/ausi-2018-0002.

- [3] I. Goodfellow, Y. Bengio, dan A. Courville, *Deep Learning*. MIT Press, 2016.
- [4] Y. LeCun, L. Bottou, Y. Bengio, dan P. Haffner, "Gradient-Based Learning Applied to Document Recognition," *Proc. IEEE*, vol. 86, no. 11, hlm. 2278–2324, 1998, doi: 10.1109/5.726791.
- [5] J. Deng dan others, "ImageNet: A Large-Scale Hierarchical Image Database," *IEEE Conf. Comput. Vis. Pattern Recognit.*, hlm. 248–255, 2009, doi: 10.1109/CVPR.2009.5206848.
- [6] TensorFlow Developers, "TensorFlow Documentation." 2024.
- [7] Keras Team, "Keras API Documentation." 2024.
- [8] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2 ed. Springer, 2022. doi: 10.1007/978-3-030-34372-9.